Abdelrahman Khalifa

Systems Engineer | Rust, C, Networking, Linux

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Hiring Team, Cloudflare
Systems Engineer, Network Protocols & Distributed Systems (London or Lisbon)

Dear Hiring Team,

I am applying for the Systems Engineer, Network Protocols & Distributed Systems role. The work it describes (moving core proxy infrastructure from NGINX to Rust, and measuring protocol performance on real traffic) is the kind of systems programming I have been doing in open source, and the area I want to go deeper in.

As a Google Summer of Code 2026 contributor with The Linux Foundation's OpenPrinting project, I built the Printers page for the COSMIC desktop in Rust. Its backend speaks IPP, an HTTP-based protocol, to CUPS, and discovers printers over DNS-SD/mDNS. That backend reaches C libraries through FFI, so I ran it as a separate varlink service inside `cosmic-settings-daemon`: a crash in native code can never take down the Settings app. To make it work, I got 7 pull requests merged upstream. I ported the cups-rs bindings to libcups3 with CI for CUPS 2 and 3, added IPP management and DNS-SD bindings, and fixed crashes and a leaked client in libcups' Avahi code, in C. I passed the final evaluation and still maintain the project.

My first open-source contribution was to the Zephyr RTOS networking stack: I implemented the `IP_MULTICAST_LOOP` socket option for IPv4 across six modules, which the maintainers reviewed and merged. I have also had changes merged in Rust Clippy and in Boa, a JavaScript engine in Rust, where I added a fast path for Latin1 strings.

I care about measuring before claiming. At EKSON, where I own the software for pneumatic motion simulators, I cut the desktop control app's CPU usage and latency by about 40%, measured, by using `asyncio` to wait for serial and network messages concurrently. I also wrote the ESP32-S3 firmware and moved it to Zephyr RTOS.

I am now studying distributed systems in depth: the Raft paper, *Designing Data-Intensive Applications* and CMU's database systems course. Next I am building a Raft-replicated key-value store in Rust. I would be glad to bring this to Cloudflare's protocols team, and I am ready to relocate to London or Lisbon.

Thank you for your time and consideration.

Sincerely,
Abdelrahman Khalifa